

AnimAlte: Designing a Multimodal AI-Infused Cartoon Video System to Support Structured Vocabulary Learning for Preschoolers at Home

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Abstract

Cartoon videos have proven effective for preschool children’s vocabulary learning, but in family contexts, insufficient structured guidance may limit children’s engagement and vocabulary transfer. This poster presents **ANIMALTE**, an AI-infused cartoon video learning system that supports structured vocabulary learning through age-appropriate vocabulary extraction, real-time visual cues with interactive Q&A, and AI-guided drawing activities for story creation. In a user study with ten families, we show that **ANIMALTE** supports structured vocabulary learning during parent–child co-viewing by enabling children to actively label and describe on-screen objects through AI-driven guided Q&A. We derive HCI design implications for AI-infused cartoon video vocabulary learning systems.

CCS Concepts

• Human-centered computing → Interactive systems and tools.

Keywords

Vocabulary learning, video-based learning, children, cartoon video, visual language model, large language model

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1 Introduction

Vocabulary learning is foundational in early childhood [10, 26], yet it is challenged by difficulties in strong context dependency [30] and insufficient opportunities for active engagement [6]. Cartoon videos are already deeply embedded in young children’s everyday media routines, leading to frequent and voluntary exposure [22]. Their rich audiovisual context provides repeated, meaningful exposure to

words that supports attention, comprehension, and long-term retention [11, 30]. Despite these benefits, learning vocabulary through cartoon videos without professional and structured guidance is unlikely to effectively facilitate efficient vocabulary acquisition [28].

To address this, Presentation-Practice-Production (PPP) is a traditional language teaching framework that structures instruction into three sequential phases: the presentation of target language by the teacher, controlled practice to reinforce accuracy, and free production that supports the learner’s independent use of the target language [8, 21]. Prior studies indicate that combining the Presentation-Practice-Production (PPP) framework with video-based learning supports vocabulary acquisition by offering contextualized visual input, guided practice, and opportunities for active word use, which together improve vocabulary retention and real-world transfer [23]. However, these instructional frameworks have been primarily applied in school contexts [8, 12], and their use in family contexts with preschool children remains challenging. Unlike trained teachers, caregivers often lack the expertise to structure the three phases to support vocabulary learning [9]. Moreover, it remains challenging for caregivers to systematically introduce or tailor age-appropriate new vocabulary from cartoon videos [16]. Further, they may struggle to formulate structured questions based on video content and to provide timely, repeated, and accurate feedback during vocabulary practice [29]. Finally, they struggle to guide children in transferring vocabulary learned from cartoons into meaningful use in real-world contexts [2, 14]. Therefore, there is a need for HCI systems that augment cartoon video viewing with AI-driven support to enable structured vocabulary learning for children in family settings; however, relatively little prior HCI work has examined how AI-enhanced cartoon videos can be designed to effectively support such structured learning.

To address these needs, we propose **ANIMALTE**, an AI-infused cartoon video system designed to support preschool children’s vocabulary learning during co-viewing with caregivers. Grounded in the Presentation-Practice-Production (PPP) framework, the system is structured into three phases to facilitate vocabulary acquisition with active caregiver engagement through introducing age-appropriate vocabulary, providing real-time visual cues, real-time Q&A and shadowing exercises during watching, and guided repetition to support practice and vocabulary consolidation.

Later in this work, we conducted a user study with ten family groups to examine how an AI-infused cartoon video learning system supports children’s vocabulary learning across the three phases. Our preliminary findings indicate that AI-driven guided



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question–answer interactions during viewing helped children move from abstract vocabulary recognition to actively labeling and describing on-screen objects through visual and verbal forms, while AI-supported drawing activities enabled children to transfer and extend vocabulary learned from videos into contexts related to their own lived experiences.

2 Related Work

Cartoon Videos for Children’s Vocabulary Learning: Cartoon videos help preschool children learn by presenting new words and ideas in engaging audiovisual stories that support attention and understanding [1]. A large body of research in education and developmental psychology has examined how cartoon videos support children’s vocabulary development through mechanisms such as repeated exposure [20], joint parent–child viewing and dialogic interaction [31], and the provision of subtitles in animated content [15]. For example, Rice et al. examined whether preschool children can learn new words from watching television by exposing them to unfamiliar vocabulary embedded in televised content, and analyzed how learning outcomes varied as a function of children’s age and word type [24]. Despite these benefits, prior work indicates that without structured learning support, cartoon/animation-based viewing alone is unlikely to yield efficient vocabulary gains for young novice learners [17, 28].

Presentation–Practice–Production in Children’s Vocabulary Learning: The Presentation–Practice–Production (PPP) approach is a structured lesson framework used in language teaching, where the target language is first presented in context, then practiced through controlled activities, and finally produced by learners in freer, communicative use [5]. Empirical studies indicate that vocabulary instruction based on the Presentation–Practice–Production (PPP) approach leads to significantly better vocabulary learning outcomes than traditional methods, with PPP-based experimental groups outperforming control groups on vocabulary tests [25]. Further, prior studies suggest that integrating video clips into a Presentation–Practice–Production-structured blended learning design is a promising approach to vocabulary instruction, supporting vocabulary acquisition, retention, and learner engagement [23]. However, these instructional frameworks have been primarily applied in school settings [8, 12], and their application in family contexts with preschool children remains limited. Unlike trained teachers, caregivers often lack the pedagogical expertise to effectively structure the three PPP phases to support vocabulary learning [9]. In addition, caregivers face difficulties in providing timely, repeated, and accurate feedback during practice [29], and guiding children to transfer vocabulary learned from cartoons into meaningful use in real-world contexts [2, 14].

AI-Supported Vocabulary Learning for Children in HCI Within the human–computer interaction community, many studies have explored how AI-driven systems support children’s vocabulary learning through personalized examples and stories [19], augmented reality–based situated learning [3], and eye-tracking based assessment of vocabulary comprehension [13]. For example, Lee et al. proposed and evaluated OSOS, a system that captures children’s everyday home language, identifies personalized target words, and uses generative AI to create custom storybooks, enabling

individualized vocabulary assessment and intervention beyond the limits of standardized methods [18]. Further, Xu et al. presented an interactive educational video system that embeds a conversational AI agent into children’s narrative videos to prompt real-time questions and personalized feedback, enabling more active vocabulary learning than passive viewing [32]. However, few studies have explored how multi-modal AI can enhance cartoon videos to support children’s vocabulary learning by providing structured guidance.

3 METHODOLOGY

Based on a formative study with three child language specialists and one speech–language pathologist (SLP), we developed **ANIMALTE**, an AI-infused cartoon video learning system that supports structured vocabulary learning for children and caregivers during co-viewing. The system helps caregivers introduce age-appropriate vocabulary from videos and provides visual cues, question–answer interactions, and guided repetition during viewing to support practice and vocabulary consolidation. It further uses AI-guided drawing activities to help children create stories with the learned words, encouraging active use in meaningful contexts and supporting vocabulary transfer.

3.1 ANIMALTE Usage Scenarios

Oli enjoys watching *Peppa Pig*. To support her vocabulary learning, her mother first used **ANIMALTE** to select age-appropriate target words and chose a *Peppa Pig* episode from the video library (Presentation)(Figure 1 A-B). The AI identified relevant vocabulary from the episode, and Oli’s mother briefly introduced these words before viewing, encouraging Oli to notice where they appeared in the cartoon (Figure 1 C). Oli then watched the episode once (Figure 1 D). In the next stage (Practice), **ANIMALTE** automatically paused the video at moments when the target words appeared and highlighted the corresponding regions through a visual cue. An AI assistant asked Oli naming and descriptive questions (e.g., “What is this?” and “Is it whole or broken?”), provided praise for correct answers, and guided her to repeat the word with feedback (Figure 1 E). After the video (Production), the AI prompted Oli to draw a story related to the learned word from her daily life (Figure 1 F). Once she finished drawing, the AI asked follow-up questions about what she drew, what happened, and where it took place (Figure 1 G), and then helped generate a story based on her drawing (Figure 1 H). These drawings and generated stories were stored within the system and could also be printed as physical artifacts, allowing children to revisit and reinforce their learning through repeated review.

3.2 ANIMALTE Pipelines and Implementation

In the Presentation phase, to support caregivers in selecting appropriate vocabulary from cartoon videos, the system first extracts time-aligned speech transcripts using Microsoft Azure Video Indexer API¹. We use GPT-4o² to analyze video transcripts and pedagogically extract candidate nouns, which are then filtered based on

¹Microsoft Azure Video Indexer API, <https://azure.microsoft.com/en-us/products/ai-video-indexer>

²GPT-4o, <https://openai.com/index/gpt-4o/>



Figure 1: User interface flow of ANIMALTE: (A)Age input, (B)Video Upload&Word Filtering, (C)Word Selected, (D)First Round Video Watching, (E)Second Round Guided Learning Watching, (F)Exploring guided painting, (G-H) Story Generation.

children’s language development stages and expert recommendations. These words are coarsely aligned with video segments using a ± 2 -second temporal window around their corresponding timestamps. Key frames sampled within this window are then matched with word semantics via GPT-4o’s multimodal understanding to determine whether each word can be effectively visualized in the scene. In the Practice stage, during viewing, we employ SAM ²³ to segment regions corresponding to the selected vocabulary in the current video frame. These segmented regions are paired with GPT-4o-generated language explanations, enabling the system to guide children’s visual attention and prompt inquiry about relevant on-screen content through structured question–answer interactions. In the Production stage, a vision large language model (GPT-4o) prompts open-ended questions about previously learned vocabulary. After the child completes the creation, the model interprets the child’s artifact and generates structured follow-up questions, which are then synthesized into a simple, child-friendly story.

4 User Study

4.1 Study Design

Participants. We recruited 10 families (children aged 3–5; 7 mothers and 3 fathers) for this exploratory study. All participants were native speakers of [Language] to ensure linguistic consistency. Informed consent was obtained from all parents in accordance with the Institutional Review Board (IRB) approval. Each family received USD 14 as compensation.

Procedure and Analysis. Each session was conducted in person in a simulated home environment. All families interacted with the same *Peppa Pig* episode and followed the same three-stage workflow using ANIMALTE. Each session lasted approximately 30 minutes, with researchers present only to provide technical support. All sessions were audio and video recorded. Afterward, we conducted 30-minute semi-structured interviews with parents about their experiences using ANIMALTE. We then performed an

inductive thematic analysis of the interview transcripts and session recordings to identify key themes and illustrative examples [4].

4.2 Results

We exemplify how the AI-infused cartoon video learning system supports children’s structured vocabulary learning during co-viewing with their parents. First, we found that **parents customized target vocabulary based on their child’s personal relevance**. Concretely, some parents selected target vocabulary based on their child’s everyday routines, familiar objects, recent events, and current interests: “The AI suggested the word ‘photo frame’. We had just taken family photos recently, so I used that word to connect the cartoon to our own photo shoot. We talked about the pictures at home, who was in them, and where they were taken. It also helped my child remember the word more easily (P3).”

Second, we found that **AI-driven guided question–answer interactions during viewing helped children move from abstract vocabulary to actively labeling and describing on-screen objects through visual and verbal forms**. For example, P7 mentioned: “Visual outlining of target words helped children understand what each word referred to and what it looked like, while incorrect responses were addressed through choice-based prompts and repeated, patient correction.” P8 also praised the AI’s patience in guiding vocabulary learning step by step: “I don’t have that much patience myself. When my child gets playful or doesn’t answer seriously, I usually can’t keep teaching. But the AI can keep encouraging him and continues guiding him step by step.”

Finally, we found that **AI-supported drawing activities helped children transfer and extend vocabulary learned from videos into different contexts related to their own lived experiences**. For instance, after the child learned the word *car*, P4 and the child co-created a drawing about driving their family’s car on a trip: “The AI asked him to draw a story related to the word ‘car’. In the cartoon, Daddy Pig was driving a red convertible to go out together, so I suggested that he draw a trip in our own family car instead (P4).” P8 further added: “It wasn’t just about drawing the words we had

²³SAM2, <https://github.com/facebookresearch/segment-anything-2>

learned. When drawing an egg, we immediately thought about the hens we have back in our rural hometown.”

5 Design Implications and Future Work

5.1 Design Implications

Implication 1: Combining visual cues with guided questioning during cartoon video viewing supports children’s vocabulary learning. Prior research has shown that pausing during video co-viewing to engage preschool children in dialogic questioning about on-screen content significantly improves three-year-olds’ story comprehension and vocabulary learning [27]. In HCI, Xu et al. (2022) showed that embedding a conversational agent in children’s educational videos to deliver contingent questions and responsive feedback increased children’s interaction and improved immediate learning outcomes compared to non-interactive videos [32]. By integrating multimodal AI, our system interprets specific regions of cartoon videos and couples them with structured questioning and personalized feedback based on children’s responses, enabling children to consolidate abstract vocabulary through visually grounded cues and verbal descriptions rather than passive viewing. Future work could further explore how AI-driven dialogic questioning can be precisely aligned with specific on-screen regions, actions, or character states. For example, future systems could reuse the same visual anchor for a learned vocabulary item across different temporal states (e.g., static vs. moving), enabling children to extend vocabulary learning from object labels to action verbs.

Implication 2: Supporting AI-driven creative production activities helps children transfer newly learned vocabulary into real-world contexts. Vocabulary learned from cartoon videos often does not easily transfer to real-life use, because the words are embedded in highly contextualized and fictional narratives, with limited direct connection to children’s everyday experiences and opportunities for practical use [2]. In the PPP framework, the Production stage uses creative activities (e.g., role-play, storytelling, drawing, or situational enactment) to help learners actively use vocabulary to construct meaning, transfer words from controlled practice to real contexts, and support vocabulary transfer and long-term retention [21]. Our system incorporates expert recommendations and uses a VLM to guide children in drawing and storytelling activities after viewing, helping them connect learned vocabulary with their own lived experiences and extend word use across different contexts. Future work could explore richer AI-driven creative production activities by connecting vocabulary learned from cartoon videos, such as guiding children to role-play with AI-controlled characters and use cartoon characters in different real-world scenarios, enabling vocabulary use across varied everyday contexts.

5.2 Future Work

First, future work will extend this study by involving a larger number of family groups and conducting long-term field evaluations. For example, we plan to invite families to use the system at home over an extended period (e.g., one month) to quantitatively assess gains in preschool children’s vocabulary learning outcomes, while qualitatively exploring how AI-mediated support shapes family vocabulary learning practices in everyday home settings. Furthermore, future work should incorporate inclusivity considerations

to better support diverse child populations in vocabulary learning during cartoon video viewing. For example, future AI-enhanced cartoon video learning systems for vocabulary acquisition should also consider children with special needs, including autistic children [7]. Such systems could support autistic children by enabling them to actively identify and click on target vocabulary items within cartoon scenes, while providing richer visual guidance. Finally, future work could quantitatively assess the effects of AI-enhanced cartoon co-viewing on family vocabulary learning by introducing comparison conditions, such as a baseline condition using structured cartoon viewing without AI support and an experimental condition incorporating AI-driven guidance.

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